GHADAI at AraGenre 2026: Definition-Guided Classification of Arabic Text Genres via Bi-Encoder Retrieval and Large Language Model Reranking

Manar Ali Alsubhi
Department of Data Science,
College of Computing,
Umm Al-Qura University,
Makkah, Saudi Arabia
manaralialsubhi1212@gmail.com

Ghader Reda Kurdi
Department of Data Science,
College of Computing,
Umm Al-Qura University,
Makkah, Saudi Arabia
grkurdi@uqu.edu.sa

Abstract

We describe the GHADAI approach to the AraGenre 2026 shared task, which requires classifying Arabic text into broad and specific genre categories, including specific genres unseen during training. Our approach is a two-stage pipeline: a fine-tuned multilingual bi-encoder scores all 27,972 test sentences against the 74 candidate definitions, and a large language model reranks the top candidates per sentence using cyclic option rotation to mitigate positional bias. On the hidden test set, the bi-encoder achieves a hierarchical macro-F1 of 0.4213, which increases to 0.5006 with Fanar-9B reranking and to 0.6050 with Qwen-32B reranking over the top 15 candidates using five rotations. The final pipeline ranked fourth on the official leaderboard. These results demonstrate the effectiveness of combining definition-based retrieval with LLM reranking for generalization to unseen Arabic text genres.

1 Introduction

Genre classification assigns a communicative category to a text, such as an analytical report, a religious commentary, or a personal narrative. Accurate genre labels benefit a wide range of downstream applications. Search and retrieval systems can filter or rank results by document type; content-moderation and routing pipelines can send legal, religious, or medical text to the appropriate specialized handler; and corpus construction for language-model training can be balanced across genres to avoid over-representing any single style. Genre is also a useful feature for downstream natural language processing tasks such as sentiment analysis, summarization, and machine translation, where the appropriate handling often depends on the text type. For Arabic in particular, where large genre-annotated resources remain scarce, a classifier that generalizes to previously unseen genres is especially valuable.

The AraGenre 2026 shared task (El-Haj et al., 2026) frames this for Arabic at two levels of granularity: a broad genre and a more specific one. The defining property of the task is that it is zero-shot by construction. The training, development, and test splits contain disjoint sets of specific genres, and at test time each genre is described only by a short English definition. A model that memorizes the mapping from surface features to a fixed label inventory cannot generalize, because the labels themselves change. This is not a hypothetical concern: on the released development leaderboard, whose gold answers ship inside the starter kit, several submissions reached 0.99 by uploading the answers, yet the same submissions collapsed to 0.68–0.72 on the hidden test set. The gap between development and test performance is the central phenomenon the task is designed to expose.

We therefore treat the genre definition as an input to every component, never as a hardcoded target. The bi-encoder learns a matching score f(text, definition) to retrieve candidate genres, while a large language model (LLM) subsequently reranks the top candidates using their definitions. Prediction consists of scoring all candidate definitions and selecting the best. This design enables the same trained pipeline to handle all previously unseen specific genres of the test set without retraining.

Our contributions are: (i) a two-stage bi-encoder-LLM reranking pipeline that scales to the 27,972-sentence test set in practical time; (ii) a cyclic-rotation scheme that mitigates positional bias in option-based LLM scoring; (iii) a model-selection rule based on worst-case performance under imbalanced resampling rather than balanced development accuracy; and (iv) observations from development experiments on prompt guidance and same-family ensembling.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 277–280
October 28-29, 2026 ©2026 Association for Computational Linguistics

2 Task and Data Description

The starter kit (training and development sets) provides 140 training sentences across seven specific genres and 110 development sentences across six specific genres, with no overlap in specific genres between the two sets. The hidden test set contains 27,972 sentences and 74 specific genres organized under six broad genres. The distribution of specific genres across broad genres is highly uneven. The Informative broad genre alone covers 42 specific genres, while Legal covers only three (Table 1). Sentences also varies considerably in sentence length, averaging 571 characters with a long tail extending to 4,530 characters.

Each split ships a definitions file mapping every specific genre to a broad genre and to English definitions for both levels (Figures 1 and 2). Because these definitions are available for the test genres, the broad-to-specific hierarchy is known at inference time and need not be inferred from the training data—a property our hierarchical experiments exploit.

File name: train

{ "id": "train_000001",

"text": " أعلنت الوزارة بدء تنفيذ المرحلة الثانية من مشروع تطوير البنية التحتية." },

File name: train_gold

{ "id": "train_000001",

"broad_genre": "Legal",

"specific_genre": "Official Statements" },

File name: train_genre_definitions

{ "broad_genre": "Creative",

"broad_genre_definition": "Expressive, reflective, or artistic writing emphasising personal voice, emotional expression, storytelling, imagination, persuasion, or stylistic creativity.",

"specific_genre": "Personal Narratives",

"specific_genre_definition": "First-person reflective or experiential texts describing memories, emotions, personal experiences, life events, or individual perspectives." },

Figure 1: Examples of the AraGenre 2026 training data. Training instances are provided with gold broad and specific genre labels.

File name: test

{ "id": "1",

"text": " يعد قطاع النقل البحري من الركائز الاساسية في الاقتصاد القومي ، فقد أدى دوراً رائداً في نقل الحضارات، وكان له تأثير في الحياة البشرية ،إذ تطورت الس فن الى ناقلات عملاقة أكثر تنوعاً وتقدماً ، وظهرت سفن الحاويات ذات الأحجام الكبيرة ،مما كان لهُ الأثر الأكبر ف عرضفن الى ناقلات عملاقة أكثر تنوعاً وتقدماً ، وظهرت سفن الحاويات ذات الأحجام الكبيرة ،مما كان لهُ الأثر الأكبر ف"},

File name: test_genre_definitions

{"broad_genre": "Creative", "broad_genre_definition": "Expressive, reflective, or artistic writing emphasising personal voice, emotional expression, storytelling, imagination, persuasion, or stylistic creativity.",

"specific_genre": "classical_poetry",

"specific_genre_definition": "Poetic writing composed in Classical Arabic using traditional literary style, formal vocabulary, rhetorical imagery, and structured poetic expression, often focusing on themes such as love, wisdom, praise, religion, or human experience." },

Figure 2: Examples of the AraGenre 2026 test data. Test instances are accompanied by candidate genre definitions.

Broad genre	# Specific genre
Informative	42
Learning	12
Creative	7
Religious	6
Interactive	4
Legal	3

Table 1: Specific-genre count per broad genre (test set).

3 Method

3.1 Definition-Guided Bi-Encoder

We fine-tune the multilingual-e5-base bi-encoder (Wang et al., 2024) on the 140 training pairs, using a sentence and the definition of its correct specific genre as a positive pair with a multiple-negatives ranking loss (Reimers and Gurevych, 2019). Arabic diacritics and tatweel are stripped; letter unification is disabled after it was found to cost 0.023 macro-F1 on development. A leave-one-genre-out evaluation gives 0.671 accuracy against a 0.143 chance rate, confirming the encoder learns genre definition matching that transfers to unseen genres rather than memorizing labels. At inference the encoder embeds all sentences and all definitions once and takes a dot product, producing a full sentence-by-definition score matrix for the 27,972-sentence test set in a few minutes. This matrix is the retrieval stage: for each sentence, it ranks the 74 candidate definitions.

3.2 LLM Reranking with Cyclic Rotation

The bi-encoder alone confuses semantically adjacent specific genres. We therefore add a rerank-

ing stage: for each sentence, the top-K definitions from the bi-encoder are presented to a LLM as lettered options, and the model scores each by the log-probability of the corresponding option letter. We use Qwen2.5 (Yang et al., 2024) and the Arabic-centric Fanar-1-9B (Abbas et al., 2025), loaded in 4-bit precision (Dettmers et al., 2023). Option-based scoring is susceptible to positional bias, whereby models may favor particular option positions independently of their content (Zheng et al., 2024; Pezeshkpour and Hruschka, 2024). The measured positional bias was 2.79 log-probability units for Qwen and 3.26 for Fanar. To mitigate this bias, we present the same K options in several cyclic rotations and sum the per-option log-probabilities across rotations. With a complete set of K rotations, each candidate occupies every position exactly once, so any additive position-specific bias cancels in the aggregate. In practice, we use a smaller number of rotations to reduce computational cost, providing partial positional averaging.

3.3 Two-Stage Retrieval and Reranking at Test Time

Scoring all 74 genres with full cyclic rotation for every sentence was estimated to require approximately 46 hours per model on the available hardware (Google Colab with an NVIDIA A100 GPU), making this approach computationally expensive and impractical within the shared-task timeline. The two-stage pipeline decouples cost from accuracy: the bi-encoder narrows 74 candidates to K per sentence, and the LLM reranks only those K. With K between 8 and 15 and three to five rotations, a full pass over the test set runs in about 12 hours. Because long jobs risk interruption, the reranker checkpoints its progress and resumes from the last completed sentence.

3.4 Model Selection

The organizers provide two baselines. The first "B1" is an embedding-based classifier that matches each sentence to the nearest genre definition in a shared embedding space, scoring 0.4658. The second "B2" is a retrieval-augmented Qwen system that prompts the model with retrieved context, scoring 0.3899. Both serve as reference points that any competitive system should exceed.

Development has only 110 sentences and six genres, so a configuration that looks best on balanced development accuracy may be overfit. We instead rank configurations by their worst-case macro-F1 across 40 Dirichlet(0.4) resamples of the development set, simulating potentially imbalanced label distributions in the test set. We select configurations based primarily on this worst-case score and prefer the simpler system when configurations show comparable performance. This criterion favors configurations that remain robust across varying label distributions rather than those optimized for the original development distribution.

System	Broad	Specific	Hier.
Organizers B1 (embedding)	—	—	0.4658
Organizers B2 (Qwen RAG)	—	—	0.3899
Bi-encoder only	0.5572	0.2853	0.4213
+ Fanar-9B rerank (top-8)	0.6913	0.3100	0.5006
+ Qwen-32B rerank (top-15)	**0.7803**	**0.4297**	**0.6050**

Table 2: Test-set macro-F1. Hierarchical macro-F1 is the mean of broad- and specific-genre scores and is the official evaluation metric.

4 Results

The evaluation metric is the hierarchical macro-F1, the mean of the broad and specific macro-F1 scores. Table 2 reports macro-F1 on the hidden test set. The organizer baselines score 0.4658 (embedding, B1) and 0.3899 (Qwen RAG, B2). The bi-encoder alone scores 0.421. Adding a Fanar-9B reranker over the top 8 candidates raises the hierarchical macro-F1 score to 0.501, driven mainly by the broad genre. Replacing the reranker with a Qwen-32B model over the top 15 candidates with five rotations raises the specific genre score from 0.31 to 0.43 and the hierarchical macro-F1 score to 0.605, which placed fourth on the official leaderboard.

The results indicate that reranking improves performance at both levels, with the largest absolute gain observed for broad-genre classification. Specific-genre classification remains more challenging, although its macro-F1 also improves substantially with the Qwen-32B reranker. The broad-

genre score also improves because the broad label is derived directly from the selected specific genre, ensuring hierarchical consistency by construction.

Two intuitive extensions failed our worst-case selection rule and were not adopted. First, adding explicit disambiguation guidance to the prompt—generic, definition-driven instructions that named no specific genre—raised balanced development accuracy but lowered worst-case macro-F1 from 0.892 to 0.842, a clear sign of development-set overfitting. Second, ensembling Fanar with the same-family SILMA-9B model provided limited diversity: the two models agreed on 91% of development predictions, and their combination underperformed the Qwen-plus-Fanar ensemble. These results suggest that model diversity may be more beneficial than simply increasing the number of models in the ensemble.

To better understand the pipeline's error patterns, we analyzed the pipeline's predictions on the training data. We found that 91% of sentences were correctly classified. Among the remaining errors, most involved an incorrect broad genre rather than an incorrect specific genre within the correct broad genre. This explains why a strict two-stage hierarchical decoder that picks the broad genre first and then the specific genre within it performed worse than our pipeline. An incorrect broad-genre decision in the first stage cannot be recovered, because the second stage is restricted to specific genres within the selected broad genre. In contrast, our pipeline ranks specific genres directly across all candidates, avoiding an early commitment to a broad genre.

5 Conclusion

We presented a definition-guided two-stage pipeline for zero-shot Arabic genre classification, combining bi-encoder retrieval with LLM reranking to generalize to specific genres unseen during training. The bi-encoder efficiently narrows the 74 candidate genres, while cyclic-rotation reranking mitigates positional bias in option-based LLM scoring. On the hidden test set, Qwen-32B reranking improved hierarchical macro-F1 from 0.4213 for the bi-encoder alone to 0.6050, placing our approach fourth on the official leaderboard. Overall, the results demonstrate the effectiveness of combining definition-based retrieval with LLM reranking for classification when the target label set is unseen during training.

Acknowledgments

The authors would like to thank the organizers of the AraGenre 2026 shared task for their efforts in organizing the task.

References

Ummar Abbas, Mohammad Shahmeer Ahmad, Firoj Alam, Enes Altinisik, Ehsannedin Asgari, Yazan Boshmaf, Sabri Boughorbel, Sanjay Chawla, Shammur Chowdhury, Fahim Dalvi, Kareem Darwish, Nadir Durrani, Mohamed Elfeky, Ahmed Elmagarmid, Mohamed Eltabakh, Masoomali Fatehkia, Anastasios Fragkopoulos, Maram Hasanain, Majd Hawasly, and 22 others. 2025. Fanar: An arabic-centric multimodal generative AI platform. *arXiv preprint arXiv:2501.13944*.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, volume 36.

Mo El-Haj, Saad Ezzini, Shadi Abudalfa, Mustafa Jarrar, Nguyen Minh Chi, and Nguyen Minh Quan. 2026. Aragenre 2026: A hierarchical definition-guided arabic genre classification shared task. In *Proceedings of the 4th Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Pouya Pezeshkpour and Estevam Hruschka. 2024. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2006–2017.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, and 23 others. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2024. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*, volume 2024, pages 19426–19454.